

HARISH ARAVINDH M

Python Full-Stack Developer | Django & React Specialist | Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Python Full-Stack Developer and Master of Computer Applications (MCA) graduate with strong expertise in architecting scalable backend microservices, RESTful APIs, and modern frontend interfaces. Proficient in Python (Django, Django REST Framework, FastAPI), JavaScript (React.js), and relational database engineering (PostgreSQL, MySQL). Demonstrated experience in building high-throughput web applications, integrating Generative AI & Retrieval-Augmented Generation (RAG) workflows, and optimizing database performance. Passionate about writing clean, maintainable code, implementing robust system design principles, and delivering production-ready enterprise software solutions.

TECHNICAL COMPETENCIES

Backend & APIs: Python (3.x), Django, Django REST Framework (DRF), FastAPI, RESTful API Architecture, Microservices, ORM, PHP

Frontend & Web: JavaScript (ES6+), React.js, HTML5, CSS3, Tailwind CSS, Bootstrap, AJAX, JSON, Responsive Web Design

Databases & Storage: PostgreSQL, MySQL, SQL Query Optimization, Database Schema Design, Database Indexing, Relational Modeling

AI / ML & GenAI: Generative AI, Large Language Models (LLMs), RAG Pipelines, Vector DBs (ChromaDB/Pinecone), Knowledge Graphs, OpenCV

DevOps & Developer Tools: Git, GitHub, Version Control, Docker (Basic), Postman, Linux/Unix, VS Code, Unit Testing (PyTest/Unittest)

Core Computer Science: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Software Architecture, SDLC, Agile/Scrum

KEY TECHNICAL PROJECTS

Generative AI Personalized E-Learning Platform | Python, Django REST Framework, React, RAG, ChromaDB

- Architected Full-Stack Application:** Designed and developed an interactive e-learning solution using Python (Django REST Framework) for backend APIs and React.js for dynamic, responsive user UI.
- Engineered RAG Pipelines:** Implemented Retrieval-Augmented Generation workflows integrating LLM orchestration, vector database indexing (ChromaDB), and knowledge graphs, accelerating context retrieval by 35%.
- Automated Document Ingestion:** Built custom NLP parsing pipelines to process unstructured learning materials, enabling context-aware semantic search and personalized study generation.
- API Optimization & Auth:** Integrated JWT authentication and designed optimized RESTful endpoints, ensuring seamless cross-origin communication and fast client response times.

Gas Station Enterprise Management System (ERP) | Python, Django, MySQL, PHP, HTML5/CSS3, JavaScript

- End-to-End System Development:** Engineered a full-stack enterprise resource planning platform featuring Role-Based Access Control (RBAC), fuel inventory auditing, and sales tracking.

- **Database Schema & Integrity:** Designed and normalized a relational MySQL database structure (3NF) handling multi-user transaction integrity with complex foreign key constraints.
- **Business Automation:** Developed automated daily stock reconciliation modules and custom reporting endpoints, improving business audit accuracy by 40% and reducing manual logging errors.

Real-Time Facial Emotion & Video Analytics System | Python, OpenCV, Data Processing Pipelines, SQL

- **Computer Vision Pipeline:** Developed a high-throughput video analytics engine using Python and OpenCV to capture real-time camera streams, isolate facial landmarks, and calculate emotion confidence levels.
- **Data Stream Automation:** Created automated background logging routines to aggregate timestamped behavioral data directly into structured database tables and analytical CSV streams.
- **Performance Optimization:** Streamlined frame processing loops and frame buffer allocations, maintaining low operational latency (60+ FPS processing) during continuous multi-stream runs.

EDUCATION

Master of Computer Applications (MCA) 2024 – 2026

Bharath Niketan College of Engineering (Affiliated with Anna University) — Theni, Tamil Nadu

Relevant Coursework: Advanced Data Structures & Algorithms, Relational Database Systems (RDBMS), Web Software Engineering, Machine Learning, System Architecture

Bachelor of Science (B.Sc.) in Computer Science 2021 – 2024

Cardamom Planters' Association (CPA) College (Affiliated with Madurai Kamaraj University) — Bodinayakanur, Tamil Nadu

Relevant Coursework: Object-Oriented Programming (Java/Python), Database Management Systems, Software Engineering Principles, Computer Networks

CERTIFICATIONS & PROFESSIONAL TRAINING

- **Professional Full-Stack Software Development Certification** — 369 Tesla (Sep 2024)
- **UI/UX Design & Frontend Prototyping Case Study** — Electronics E-Commerce Platform (Jul 2025)

LANGUAGES

English (Professional Working Proficiency) | Tamil (Native / Bilingual)